

programmers, I am planning to cover data science in this column over the next few months.

In the next few columns, we will be looking at various topics in information retrieval, machine learning, data mining and natural language processing with a specific focus on the programming tools and algorithmic techniques in these areas. I plan to discuss R programming language, PANDAS (python data mining), NLTK (Python natural language tool kit) and SciKit. Data science also requires us to refresh some of our mathematical knowledge in probability and statistics and, hence, we would also be discussing the essential mathematical background needed for data science.

This month's technical interview questions

1. In a kernel module that you have written, a spin lock is getting acquired. When the lock is held, an interrupt comes in, which is getting serviced on the same processor that your kernel module is running on. Now, if the interrupt handler tries to acquire the same spin lock, what would happen? How can this situation be avoided?
2. There are a number of processes executing on the Linux system. When an out of memory condition is encountered, the kernel makes the decision to kill one or more of the processes to free up memory. How would you ensure that your process does not get targeted for a 'kill' by the 'OOM' handler?
3. The 'mkfs' command is used to build a Linux file system on a device, typically a hard-disk partition. Now, while the 'mkfs' command is running, you made the underlying hard disk 'write-protected'. What do you expect will happen?
4. In the Linux kernel, what is CPU hard lock-up and what is CPU soft lock-up? How does the Linux kernel detect these situations? When are such situations encountered?
5. What are lockless data structures? How would you implement a lockless list that supports the *insert*, *delete* and *search* operations from multiple threads?
6. What is meant by the voltage frequency scaling of processors? Where is it used?
7. Can you explain what exactly happens when you type a search keyword in the Google search bar? Please give a step by step description of all the events that occur till the search results are returned?
8. Can you explain how an anti-virus scanner works?
9. What is the difference between 'allocate on write' and 'update in place' file systems? Can you give an example for each?
10. You are given access to a Linux kernel environment in which you need to run your application. You need to write a script/program to determine whether your application is running inside a Linux KVM (kernel virtual machine). How would you do this?

My 'must-read book' for this month

This month's suggestion comes from one of our readers, Ramya, whose recommendation is the book, 'Chances are — Adventures in Probability' by Michael Kaplan and Ellen Kaplan. According to Ramya, "Though this is not a computer science book, a firm foundation in probability is important for all computer science students and this is one of the best books for a layman's introduction to probability." Thank you, Ramya, for your recommendation.

If you have a favourite programming book/article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful so I can mention it in the column. This would help many readers who want to improve their software skills.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming and here's wishing you a great and happy 2014! 

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The author is an expert in systems software and is currently working with Hewlett Packard India Ltd. Her interests include compilers, multi-core and storage systems. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group, 'Computer Science Interview Training India' at [http://www.linkedin.com/groups?home=HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"&HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"&gid=2339182](http://www.linkedin.com/groups?home=HYPERLINK%20http://www.linkedin.com/groups?home=&gid=2339182%20HYPERLINK%20http://www.linkedin.com/groups?home=&gid=2339182%20gid=2339182)

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